

Machine Learning Datasets and Model Selection

Contents

1	Introduction	3
2	Tabular Data	3
2.1	Definition	3
2.2	Classical Models	4
2.3	Deep Learning Models	4
2.4	Model Selection	5
3	Time-Series Data	5
3.1	Definition	5
3.2	Classical Statistical Models	5
3.3	Deep Learning Models	6
3.4	Model Selection	6
4	Spatial Grid Data	6
5	Image Data	7
5.1	Classical Models	7
5.2	Deep Learning	7
5.3	Model Selection	7
6	Video Data	8
7	Text and Sequential Language Data	8
7.1	Classical Models	8
7.2	Deep Learning	9
8	Audio and Speech Data	9
9	Graph Data	10
9.1	Classical Methods	10
9.2	Deep Learning	10
10	Point-Cloud Data	11
11	Set-Valued Data	11
12	Functional Data	12
13	Spatiotemporal Data	12
14	Scientific Simulation Data	13

15 Neural Operators	14
15.1 Fourier Neural Operator	14
15.2 DeepONet	14
15.3 Graph Neural Operators	14
16 3D Volumetric Data	15
17 Multimodal Data	15
18 Categorical Data	16
19 Imbalanced Datasets	16
20 High-Dimensional Data	17
21 Small-Dataset Regime	17
22 Large-Scale Datasets	18
23 Unsupervised Datasets	18
23.1 Classical Models	18
23.2 Deep Learning	19
24 Generative Data	19
25 Reinforcement Learning Data	19
26 Dataset Structure and Model Selection	20
27 The Most Important Principle	21
28 Classical versus Deep Learning	22
29 Scientific Machine Learning Model Selection	22
30 Final Model-Selection Guide	23

1 Introduction

Machine learning models do not operate on a single universal type of data. The mathematical structure of the dataset strongly influences which model is appropriate.

A dataset may consist of

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N, \quad (1)$$

where x_i is an input and y_i is the corresponding target.

However, the structure of x_i can vary substantially:

$$x_i \in \begin{cases} \mathbb{R}^d & \text{tabular data,} \\ \mathbb{R}^{T \times d} & \text{time series,} \\ \mathbb{R}^{H \times W \times C} & \text{images,} \\ \mathbb{R}^{T \times H \times W \times C} & \text{video,} \\ \mathbb{R}^L & \text{text sequences,} \\ \mathbb{R}^{N \times 3} & \text{point clouds,} \\ G = (V, E) & \text{graphs,} \\ u : \Omega \rightarrow \mathbb{R}^m & \text{fields/functions.} \end{cases} \quad (2)$$

The choice of model should therefore follow the structure of the data rather than simply the size of the neural network.

This chapter classifies the principal dataset types encountered in machine learning and compares suitable classical and deep-learning models.

2 Tabular Data

2.1 Definition

Tabular data consists of observations represented by a fixed number of features.

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \cdots & x_{Nd} \end{bmatrix}. \quad (3)$$

Each row corresponds to an observation and each column corresponds to a feature. Examples include:

- financial datasets,
- medical records,
- customer data,
- credit-risk datasets,
- experimental measurements,
- structured scientific datasets.

Features may be continuous,

$$x_j \in \mathbb{R}, \quad (4)$$

categorical,

$$x_j \in \{c_1, \dots, c_k\}, \tag{5}$$

ordinal, binary, or count-valued.

2.2 Classical Models

Important classical models include:

- Linear Regression,
- Logistic Regression,
- Ridge Regression,
- Lasso,
- Elastic Net,
- Linear Discriminant Analysis,
- Quadratic Discriminant Analysis,
- Naive Bayes,
- k -Nearest Neighbors,
- Decision Trees,
- Random Forests,
- Gradient Boosting,
- Support Vector Machines.

For nonlinear tabular data, tree ensembles are frequently among the strongest classical choices.

2.3 Deep Learning Models

Deep-learning alternatives include

- Multilayer Perceptrons (MLPs),
- TabNet,
- FT-Transformer,
- TabTransformer,
- NODE,
- Deep & Cross Networks.

For ordinary medium-sized tabular datasets, an MLP is not automatically better than a tree ensemble.

2.4 Model Selection

Dataset characteristic	Strong models	Reason
Small, linear	Linear/Ridge/Lasso	Low variance
Small, nonlinear	SVM/RF	Good sample efficiency
Mixed features	Gradient Boosting	Handles heterogeneous features
Large tabular	Boosting/MLP	Scales well
Very high-dimensional	Linear/SVM/Boosting	Regularization
Categorical-heavy	CatBoost/Boosting	Strong categorical handling

3 Time-Series Data

3.1 Definition

Time-series data contains observations ordered in time:

$$\{x_t\}_{t=1}^T. \quad (6)$$

For multivariate time series,

$$x_t = \begin{bmatrix} x_t^{(1)} & x_t^{(2)} & \dots & x_t^{(d)} \end{bmatrix}^T. \quad (7)$$

The defining characteristic is temporal dependence:

$$p(x_t) \neq p(x_t | x_{t-1}) \quad (8)$$

in general.

More generally,

$$x_t = f(x_{t-1}, x_{t-2}, \dots) + \epsilon_t. \quad (9)$$

3.2 Classical Statistical Models

Important models include:

- Autoregressive (AR) models,
- Moving Average (MA) models,
- ARMA,
- ARIMA,
- SARIMA,
- VAR,
- Exponential Smoothing,
- State-Space Models,
- Kalman Filters,
- Gaussian Processes.

An $\text{AR}(p)$ model is

$$x_t = c + \sum_{i=1}^p \phi_i x_{t-i} + \epsilon_t. \quad (10)$$

3.3 Deep Learning Models

Important architectures include:

- RNN,
- LSTM,
- GRU,
- Temporal CNN,
- TCN,
- Transformer,
- Temporal Fusion Transformer,
- PatchTST,
- TimesNet,
- N-BEATS,
- N-HiTS.

3.4 Model Selection

Problem	Strong choices	Typical advantage
Short, simple series	ARIMA/ETS	Interpretable
Multivariate dynamics	VAR/State Space	Explicit dependence
Long sequences	TCN/Transformer	Long-range structure
Small dataset	Statistical models	Sample efficiency
Large dataset	Transformer/PatchTST	Representation learning
Hierarchical forecasting	N-BEATS/N-HiTS	Strong forecasting structure

4 Spatial Grid Data

Spatial grid data is represented on a regular grid.

For a two-dimensional scalar field,

$$u = u(x, y), \quad (11)$$

and its discretization is

$$U \in \mathbb{R}^{H \times W}. \quad (12)$$

Examples include:

- images,
- temperature fields,
- pressure fields,
- fluid velocity fields,
- weather maps,
- medical imaging.

5 Image Data

An image is typically represented as

$$X \in \mathbb{R}^{H \times W \times C}, \quad (13)$$

where H is height, W is width, and C is the number of channels.

5.1 Classical Models

Classical approaches include:

- HOG + SVM,
- SIFT + Bag-of-Features,
- PCA + classifier,
- Random Forest,
- nearest-neighbor methods.

These methods generally require explicit feature engineering.

5.2 Deep Learning

Important models include:

- CNN,
- ResNet,
- DenseNet,
- EfficientNet,
- Vision Transformer (ViT),
- Swin Transformer,
- ConvNeXt.

A convolution can be represented as

$$y_{i,j} = \sum_m \sum_n K_{m,n} x_{i+m,j+n}. \quad (14)$$

Convolution exploits local spatial structure and translation-related patterns.

5.3 Model Selection

Image condition	Preferred models	Reason
Small dataset	Transfer learning/CNN	Efficient use of data
Large dataset	CNN/ViT	High representation capacity
Local structure	CNN/ResNet	Inductive bias
Long-range interactions	ViT/Swin	Global attention
Limited computation	EfficientNet/ConvNeXt	Efficiency

6 Video Data

Video introduces a temporal dimension to image data:

$$X \in \mathbb{R}^{T \times H \times W \times C}. \quad (15)$$

The model must learn both spatial and temporal dependencies.

Classical approaches include optical-flow-based methods and hand-engineered motion descriptors.

Deep-learning approaches include:

- 3D CNNs,
- ConvLSTM,
- CNN-LSTM,
- Video Transformers,
- TimeSformer,
- Video Swin Transformer.

7 Text and Sequential Language Data

Text consists of discrete symbols or tokens.

A sentence can be represented as

$$X = (x_1, x_2, \dots, x_L). \quad (16)$$

Each token is mapped to an embedding,

$$e_i = E(x_i) \in \mathbb{R}^d. \quad (17)$$

7.1 Classical Models

Classical approaches include:

- Bag-of-Words,
- TF-IDF,
- Naive Bayes,
- Logistic Regression,
- Linear SVM,
- Latent Semantic Analysis.

7.2 Deep Learning

Important models include:

- RNN,
- LSTM,
- GRU,
- CNN for text,
- Transformer,
- BERT-style encoders,
- autoregressive Transformer models.

Transformer attention is

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V. \quad (18)$$

For modern large-scale language tasks, Transformer architectures are the dominant deep-learning paradigm.

8 Audio and Speech Data

Audio is naturally a temporal signal,

$$x(t). \quad (19)$$

A discrete recording is

$$x[n], \quad n = 1, \dots, N. \quad (20)$$

A common representation is the spectrogram,

$$S(\tau, \omega) = \left| \sum_n x[n] w[n - \tau] e^{-i\omega n} \right|^2. \quad (21)$$

Models include:

- HMM,
- GMM,
- MFCC + classical classifier,
- CNN,
- RNN/LSTM,
- Conformer,
- Audio Transformer.

For speech recognition, modern Transformer and Conformer models are generally preferred when sufficient training data are available.

9 Graph Data

A graph is

$$G = (V, E), \quad (22)$$

where V is the set of nodes and E is the set of edges.

The adjacency matrix is

$$A_{ij} = \begin{cases} 1 & (i, j) \in E, \\ 0 & \text{otherwise.} \end{cases} \quad (23)$$

Graphs are useful for:

- molecular structures,
- social networks,
- traffic networks,
- citation networks,
- physical interaction systems.

9.1 Classical Methods

Classical graph methods include:

- spectral methods,
- graph kernels,
- PageRank,
- graph Laplacian methods,
- label propagation.

9.2 Deep Learning

Important models include:

- GCN,
- GraphSAGE,
- GAT,
- GIN,
- Message Passing Neural Networks,
- Graph Transformers.

A generic message-passing layer is

$$m_i^{(l)} = \text{AGG} \left\{ \phi \left(h_i^{(l)}, h_j^{(l)}, e_{ij} \right) : j \in \mathcal{N}(i) \right\}, \quad (24)$$

followed by

$$h_i^{(l+1)} = \psi \left(h_i^{(l)}, m_i^{(l)} \right). \quad (25)$$

10 Point-Cloud Data

A point cloud is an unordered collection

$$X = \{x_i\}_{i=1}^N, \quad x_i \in \mathbb{R}^d. \quad (26)$$

Unlike an image, there is no fixed grid ordering.

Applications include:

- LiDAR,
- 3D reconstruction,
- molecular geometry,
- robotics,
- computational geometry.

Important deep-learning models include:

- PointNet,
- PointNet++,
- DGCNN,
- Point Transformer,
- Point-Voxel methods.

PointNet uses a permutation-invariant aggregation,

$$f(X) = \gamma \left(\max_{i=1, \dots, N} \phi(x_i) \right). \quad (27)$$

11 Set-Valued Data

Some datasets consist of unordered sets rather than sequences.

$$X = \{x_1, x_2, \dots, x_N\}. \quad (28)$$

The output should satisfy permutation invariance,

$$f(X) = f(\pi(X)) \quad (29)$$

for any permutation π .

Appropriate models include:

- Deep Sets,
- Set Transformers,
- attention-based set models.

A Deep Sets architecture has the form

$$f(X) = \rho \left(\sum_{x \in X} \phi(x) \right). \quad (30)$$

12 Functional Data

Functional data consists of observations that are themselves functions.

$$x_i = f_i(t), \quad t \in \Omega. \quad (31)$$

The dataset is therefore

$$\mathcal{D} = \{f_i\}_{i=1}^N. \quad (32)$$

Examples include:

- temperature curves,
- spectroscopy,
- experimental trajectories,
- concentration profiles,
- physical fields.

Classical methods include:

- Functional PCA,
- spline regression,
- Gaussian processes,
- functional linear regression.

Deep-learning methods include:

- neural basis expansions,
- DeepONet,
- neural operators,
- operator Transformers.

13 Spatiotemporal Data

Spatiotemporal data contains both spatial and temporal dimensions:

$$u(x, t). \quad (33)$$

A discretized representation may be

$$U \in \mathbb{R}^{T \times H \times W \times C}. \quad (34)$$

Examples include:

- weather,
- climate,
- fluid dynamics,

- traffic,
- ocean dynamics,
- atmospheric dynamics.

Models include:

- ConvLSTM,
- 3D CNN,
- CNN + Transformer,
- Spatiotemporal GNN,
- Neural Operators,
- Fourier Neural Operators,
- PDE-informed neural operators.

14 Scientific Simulation Data

Scientific datasets frequently consist of numerical solutions of differential equations.

For example,

$$\frac{\partial u}{\partial t} = \mathcal{N}(u; \lambda), \quad (35)$$

may generate a dataset

$$\mathcal{D} = \{(a_i, u_i)\}_{i=1}^N. \quad (36)$$

Here a_i can represent

- PDE coefficients,
- initial conditions,
- boundary conditions,
- forcing functions,
- material parameters,
- geometry.

The target is often an entire field rather than a scalar.

This distinction is important.

A conventional regression model learns

$$x \longrightarrow y, \quad (37)$$

whereas an operator-learning model learns

$$a(\cdot) \longrightarrow u(\cdot). \quad (38)$$

15 Neural Operators

Neural operators are designed for mappings between function spaces,

$$\mathcal{G} : \mathcal{A} \rightarrow \mathcal{U}. \quad (39)$$

Examples include:

- DeepONet,
- FNO,
- GNO,
- GNOT,
- Transolver,
- PINO-based architectures.

15.1 Fourier Neural Operator

The FNO applies a Fourier transformation,

$$\hat{u} = \mathcal{F}(u), \quad (40)$$

performs a learned spectral transformation,

$$\hat{v}_k = W_k \hat{u}_k, \quad (41)$$

and transforms back,

$$v = \mathcal{F}^{-1}(\hat{v}). \quad (42)$$

FNOs are particularly effective for PDE solution fields on regular grids.

15.2 DeepONet

DeepONet decomposes an operator using branch and trunk networks:

$$\mathcal{G}(a)(y) = \sum_{k=1}^p b_k(a) t_k(y). \quad (43)$$

It is particularly useful when the input is a function and the output must be evaluated at arbitrary coordinates.

15.3 Graph Neural Operators

Graph-based neural operators are suitable for irregular spatial domains and meshes.

$$\mathcal{G} : \mathcal{F}(\Omega) \rightarrow \mathcal{F}(\Omega). \quad (44)$$

They are particularly useful when the geometry cannot naturally be represented by a rectangular grid.

16 3D Volumetric Data

Three-dimensional fields have the form

$$u(x, y, z). \quad (45)$$

A discretization is

$$U \in \mathbb{R}^{D \times H \times W \times C}. \quad (46)$$

Examples include:

- CT and MRI,
- computational fluid dynamics,
- 3D materials,
- molecular density fields,
- seismic volumes.

Models include:

- 3D CNNs,
- 3D U-Net,
- Vision Transformers,
- Swin Transformers,
- 3D neural operators.

17 Multimodal Data

Multimodal datasets contain multiple data modalities.

For example,

$$X = (X_{\text{text}}, X_{\text{image}}, X_{\text{audio}}). \quad (47)$$

Each modality has different structure.

Possible architectures include:

- multimodal Transformers,
- cross-attention models,
- CLIP-style contrastive models,
- multimodal foundation models.

Cross-attention can be written as

$$\text{CrossAttention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V. \quad (48)$$

18 Categorical Data

Categorical variables take values from a finite set,

$$x \in \{c_1, \dots, c_K\}. \quad (49)$$

They can be represented using one-hot encoding,

$$c_j \longrightarrow e_j \in \mathbb{R}^K, \quad (50)$$

or learned embeddings,

$$e_j = E[c_j]. \quad (51)$$

Tree-based models are particularly strong for categorical tabular data, while embedding-based neural models are useful for very large datasets.

19 Imbalanced Datasets

In classification, an imbalanced dataset satisfies

$$P(Y = c_1) \gg P(Y = c_2). \quad (52)$$

For example, fraud detection may contain very few positive examples. Accuracy becomes misleading:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \quad (53)$$

More informative metrics include

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (54)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (55)$$

and

$$F_1 = \frac{2PR}{P + R}. \quad (56)$$

Suitable approaches include:

- class-weighted logistic regression,
- class-weighted SVM,
- balanced Random Forest,
- gradient boosting,
- focal-loss neural networks.

20 High-Dimensional Data

A dataset is high-dimensional when

$$d \gg N \tag{57}$$

or when d is sufficiently large relative to the available samples.

Examples include:

- genomics,
- spectroscopy,
- text,
- hyperspectral imaging,
- high-dimensional scientific measurements.

Useful methods include:

- PCA,
- sparse regression,
- Lasso,
- Elastic Net,
- SVM,
- regularized neural networks,
- autoencoders.

PCA solves

$$\max_{\|w\|=1} \text{Var}(Xw). \tag{58}$$

21 Small-Dataset Regime

When

$$N \text{ is small,} \tag{59}$$

high-capacity neural networks can overfit.

Classical models are often preferable:

- Linear/Ridge/Lasso regression,
- Logistic Regression,
- SVM,
- Gaussian Processes,
- Random Forest,

- Gradient Boosting.

The bias-variance trade-off is

$$\mathbb{E} \left[(y - \hat{f}(x))^2 \right] = \text{Bias}^2 + \text{Variance} + \sigma^2. \quad (60)$$

Deep learning becomes more attractive as the amount and complexity of data increase.

22 Large-Scale Datasets

Large datasets allow high-capacity models to learn complex representations.

Examples include

- large image datasets,
- internet-scale text,
- large speech corpora,
- massive multimodal datasets.

Suitable models include:

- deep CNNs,
- Transformers,
- Vision Transformers,
- large language models,
- multimodal Transformers.

The advantage comes from the ability to learn representations directly from raw data rather than relying entirely on manually engineered features.

23 Unsupervised Datasets

In unsupervised learning,

$$\mathcal{D} = \{x_i\}_{i=1}^N \quad (61)$$

contains no target variable.

23.1 Classical Models

Important approaches include:

- k -means,
- Gaussian Mixture Models,
- PCA,
- DBSCAN,
- hierarchical clustering,

- Kernel PCA.

A Gaussian mixture model represents

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k). \quad (62)$$

23.2 Deep Learning

Important models include:

- Autoencoders,
- Variational Autoencoders,
- self-supervised Transformers,
- contrastive learning models,
- diffusion models.

24 Generative Data

Generative modelling attempts to learn

$$p_{\theta}(x). \quad (63)$$

Classical generative approaches include

- Gaussian Mixture Models,
- Kernel Density Estimation,
- Bayesian models,
- Hidden Markov Models.

Deep generative models include

- Variational Autoencoders,
- Generative Adversarial Networks,
- autoregressive Transformers,
- diffusion models.

25 Reinforcement Learning Data

Reinforcement learning does not normally begin with a fixed supervised dataset.

The agent observes a state

$$s_t, \quad (64)$$

takes an action

$$a_t, \tag{65}$$

receives reward

$$r_t, \tag{66}$$

and transitions to

$$s_{t+1}. \tag{67}$$

The trajectory is

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots). \tag{68}$$

Classical approaches include:

- Q-learning,
- SARSA,
- Dynamic Programming,
- Monte Carlo methods.

Deep reinforcement-learning methods include:

- DQN,
- DDPG,
- TD3,
- SAC,
- PPO,
- Actor-Critic methods.

26 Dataset Structure and Model Selection

The following table summarizes the principal relationships.

Dataset	Classical Models	Deep Models
Tabular	Logistic Regression, SVM, RF, Gradient Boosting	MLP, TabNet, FT-Transformer
Time series	ARIMA, VAR, State Space, Kalman	LSTM, TCN, Transformer, PatchTST
Images	HOG+SVM, SIFT	CNN, ResNet, ViT, Swin
Video	Optical flow methods	3D CNN, Video Transformer
Text	TF-IDF + SVM, Naive Bayes	Transformer, BERT, LLMs
Audio	HMM, GMM	CNN, Conformer, Transformer

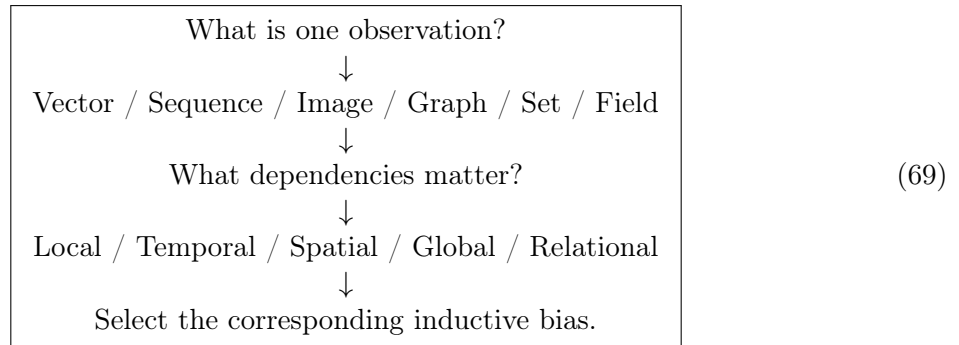
Graphs		Graph kernels, spectral methods	GCN, GAT, GIN, Graph Transformer
Point clouds		Geometric descriptors	PointNet, DGCNN, Point Transformer
Sets		Kernel methods	Deep Sets, Set Transformer
Functional data		FPCA, GP, spline methods	DeepONet, Neural Operators
Spatial fields		Numerical/statistical methods	CNN, U-Net, Neural Operators
Spatiotemporal fields		State-space models	ConvLSTM, Transformer, Neural Operators
Scientific data	PDE	Reduced-order models, GP	FNO, DeepONet, GNO, Transolver, PINO
3D volumes		PCA, classical image methods	3D CNN, 3D U-Net, 3D ViT
Unsupervised		PCA, GMM, k -means, DBSCAN	AE, VAE, contrastive learning
Generative		GMM, KDE, HMM	VAE, GAN, Transformer, diffusion
RL trajectories		Q-learning, SARSA	DQN, PPO, SAC, TD3

27 The Most Important Principle

There is no universally best machine-learning model.

Model selection should be based on the mathematical structure of the dataset.

A useful hierarchy is



For example,

$$\text{grid} \rightarrow \text{CNN/FNO}, \quad (70)$$

$$\text{sequence} \rightarrow \text{RNN/TCN/Transformer}, \quad (71)$$

$$\text{graph} \rightarrow \text{GNN}, \quad (72)$$

$$\text{point cloud} \rightarrow \text{PointNet/Point Transformer}, \quad (73)$$

and

$$\text{function} \rightarrow \text{DeepONet/Neural Operator.} \quad (74)$$

28 Classical versus Deep Learning

Classical machine learning generally performs better when the dataset is small, structured, and represented by meaningful engineered features.

Deep learning becomes increasingly attractive when

$$N \rightarrow \text{large}, \quad (75)$$

the input is high-dimensional,

$$d \rightarrow \text{large}, \quad (76)$$

and the raw data contains complex spatial, temporal, relational, or hierarchical structure. A simplified comparison is

	Classical ML	Deep Learning
Small data	Usually strong	Often overfits
Large data	Saturates earlier	Often improves
Feature engineering	Important	Often reduced
Interpretability	Often higher	Often lower
Images	Limited	Excellent
Raw text	Limited	Excellent
Graphs	Specialized methods	GNNs
PDE fields	Limited	Neural operators
Training cost	Usually lower	Usually higher

29 Scientific Machine Learning Model Selection

For scientific computing, the distinction between ordinary deep learning and neural operators is particularly important.

If the input is a finite-dimensional vector,

$$x \in \mathbb{R}^d, \quad (77)$$

ordinary regression is appropriate.

If the input is a time series,

$$x(t), \quad (78)$$

temporal models are appropriate.

If the input is an image or regular spatial field,

$$u(x, y), \quad (79)$$

CNNs or vision Transformers may be appropriate.

If the task is to learn a mapping between fields,

$$a(x) \mapsto u(x), \quad (80)$$

neural operators become particularly appropriate.

If the mapping must additionally satisfy a known PDE,

$$\mathcal{F}[u; a] = 0, \quad (81)$$

a physics-informed neural operator can be constructed:

$$\boxed{\text{Neural Operator} + \text{Physics Residual} = \text{PINO}.} \quad (82)$$

30 Final Model-Selection Guide

Data structure	First models to consider	
Small tabular	Linear/SVM/Random Forest	
Large tabular	Gradient Boosting/MLP	
Time series	ARIMA/TCN/Transformer	
Image	CNN/ViT	
Video	3D CNN/Video Transformer	
Text	Transformer	
Audio	CNN/Conformer	
Graph	GNN/Graph Transformer	
Point cloud	PointNet/Point Transformer	(83)
Set	Deep Sets/Set Transformer	
Function	DeepONet/Neural Operator	
Regular-grid PDE	FNO	
Irregular PDE geometry	GNO/Graph-based operators	
General PDE geometry	Transolver/related operators	
PDE + physical constraints	PINO	
Unsupervised vectors	PCA/GMM/k-means	
Generative	VAE/GAN/Diffusion/Transformer	
RL trajectories	Q-learning/PPO/SAC	